CS 277: AI & Mental Health Project

Text + VA: A two-dimensional foundational approach to prediction of depression

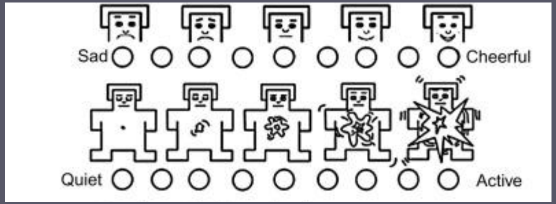
[christoh], [ion2004], [yalcintr], [kevinaw], [mikor]

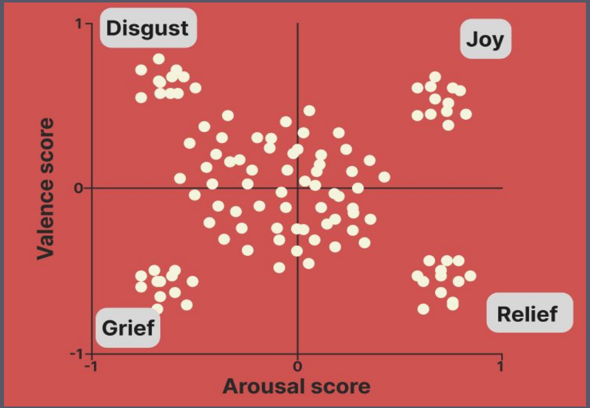
Need: Scalable clinician capacity

Current approaches:
- Agents built on domain specific models [1]
- Mental health assessment is archaic [2]
- Models measure siloed emotion [3]

Our insights:
- Foundational metric + text → mental state
- Approach could be deployed continuously
- Valence Arousal = 2D emotional scale

Solution: Valence-arousal as a psychometric signal





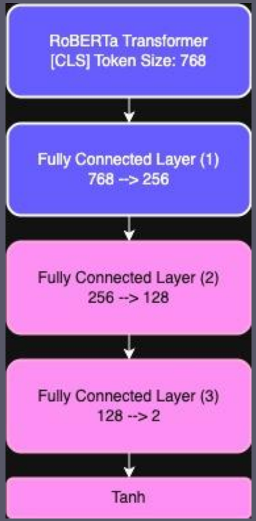
[1]: https://doi.org/10.48550/arXiv.2110.15621
[2]: https://doi.org/10.1002/wps.21191
[3]: https://arxiv.org/html/2408.13718v1

1. Predicting Valence-Arousal (VA) from Text

GPT o4 Mini

2D MAE: 0.197

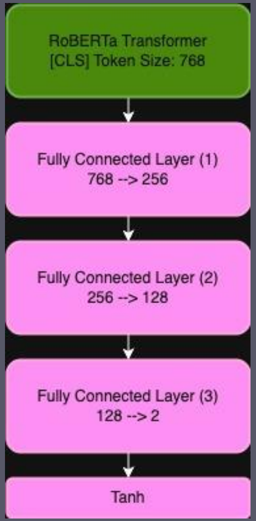
GoEmotions + Emobank[1]



2D Val MAE before Unfreezing: 0.0946

2D Test MAE: 0.0792

Emobank



2D Val MAE before Unfreezing: 0.0917

2D Test MAE: 0.0759

[1] https://github.com/kashyaparun25/Mental-Health-Chatbot-using-RoBERTa-and-Gemini/blob/main/README.md

2. Prediction of depressive patients

PHQ-8 as a metric for self-reported depression

- A patient questionnaire that corresponds to criteria for depression diagnosis (per DSM-5)
- Ranges from 0-24, higher scores indicating more severe symptoms of depression
- Used clinically for patient screening

DAIC-WOZ as a dataset for text-to-depression prediction

- Clinical interviews (average 16 minutes) of 189 people
- Interviews paired with self-reported PHQ-8 score of patients
- DAIC-WOZ has been widely used for benchmark setting using published models

Over the last *2 weeks*, how often have you been bothered by any of the following problems?
(use "✓" to indicate your answer)

	Not at all	Several days	More than half the days	Nearly every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3
3. Trouble falling or staying asleep, or sleeping too much	0	1	2	3
4. Feeling tired or having little energy	0	1	2	3
5. Poor appetite or overeating	0	1	2	3
6. Feeling bad about yourself—or that you are a failure or have let yourself or your family down	0	1	2	3
7. Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
8. Moving or speaking so slowly that other people could have noticed. Or the opposite — being so figety or restless that you have been moving around a lot more than usual	0	1	2	3
9. Thoughts that you would be better off dead, or of hurting yourself	0	1	2	3
add columns		+	+	
(Healthcare professional: For interpretation of TOTAL, please refer to accompanying scoring card). TOTAL:				

[3] Stanford University School of Medicine, *Patient Health Questionnaire (PHQ-9)*

2. Prediction of depressive patients

Hypothesis:

Labelling DAIC-WOZ transcripts with their valence-arousal scores could improve depression prediction

Our approach:

DAIC-WOZ with or without predicted VA → 3 models →

- Base GPT-o4-mini
- GPT-o4-mini with autoCOT prompting
- GPT-o4-mini with supervised fine-tuning

→ Predicted PHQ-8 score

Evaluation against true PHQ-8 using Mean Absolute Error (MAE)

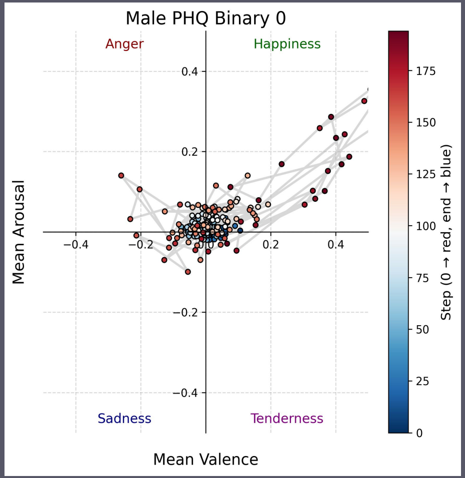
	Base GPT-o4-mini	**GPT-o4-mini with autoCOT prompting**	**GPT-o4-mini with SFT**
DAIC-WOZ transcripts	4.00	3.76	3.66
DAIC-WOZ transcripts + predicted VA	3.86	3.75	3.21

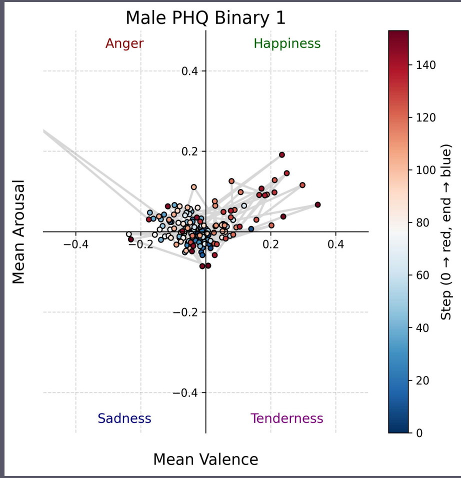
Published models

Source	**Model**	**MAE (PHQ-8)**
Lau et al 2023	BERT with prefix-tuning	3.80
Milinsenvich et al 2023	RoBERTa + hierarchical regression	3.78
Sadeghi et al 2023	GPT 3.4 + DepRoBERTa FT	3.65

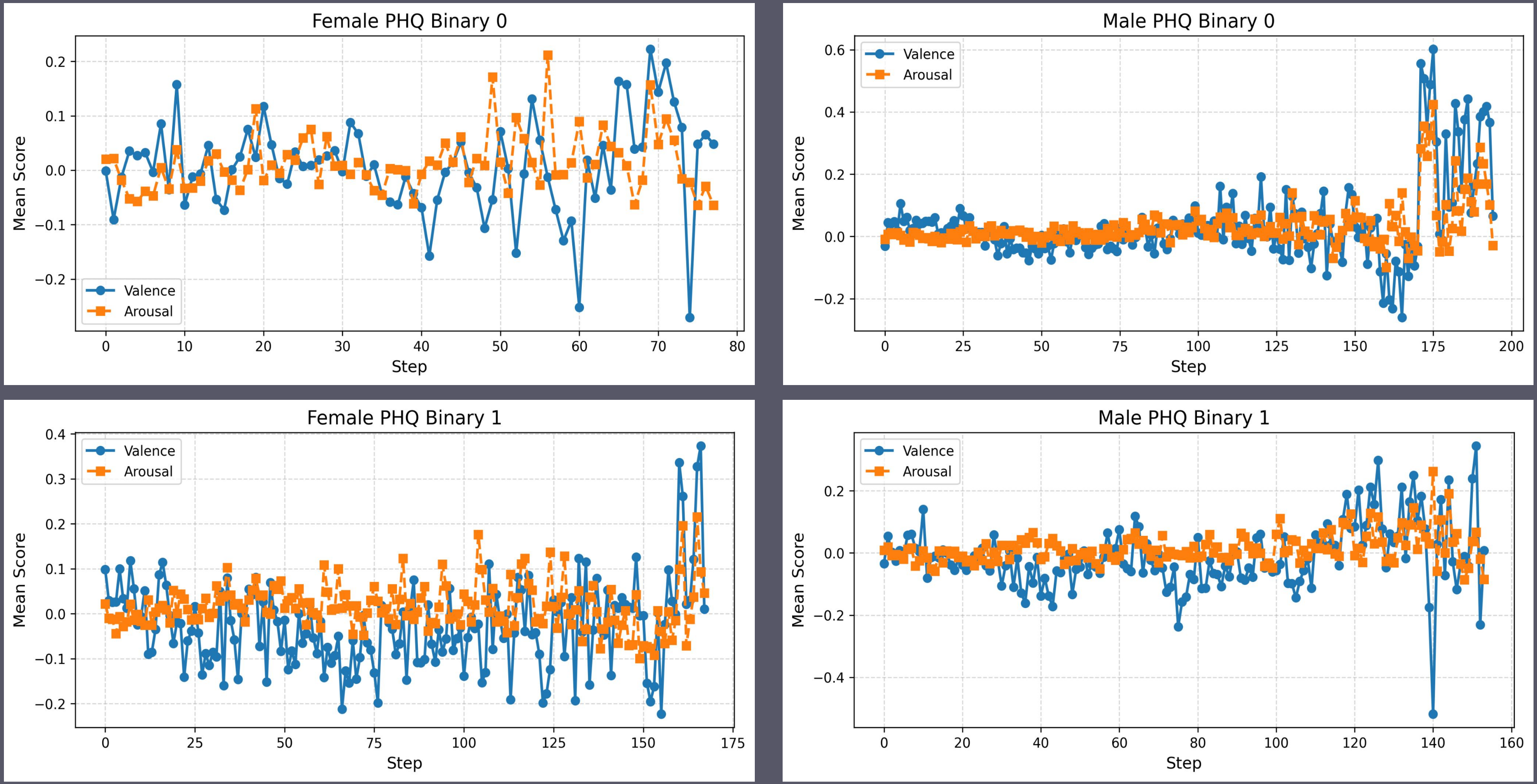
Next Steps

- **Additional Validation:** Trajectories of VA confirm relevance of depression
- **Domain-Specific:** Likely best domain-specific model to experiment with: GPT-o4-mini + autoCOT prompting + SFT
- **Foundational approach:** Investigate utility of VA for other (eg: PTSD) score prediction (also available in DIAC-WOZ)
- **Explore another quantitative affective metric:** i.e., heightened alert for PTSD
  - With enough metrics, we can hopefully build an AI clinician

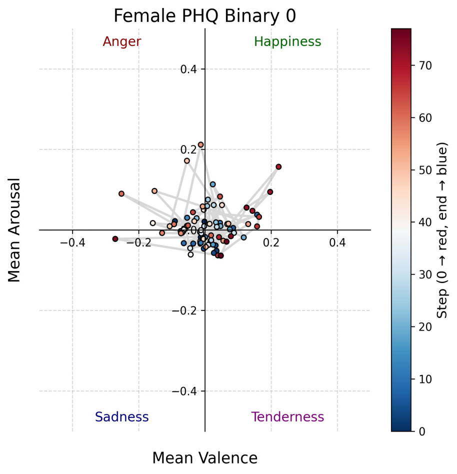




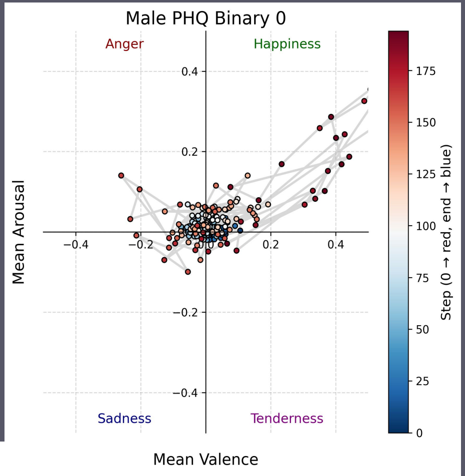
Trash can following

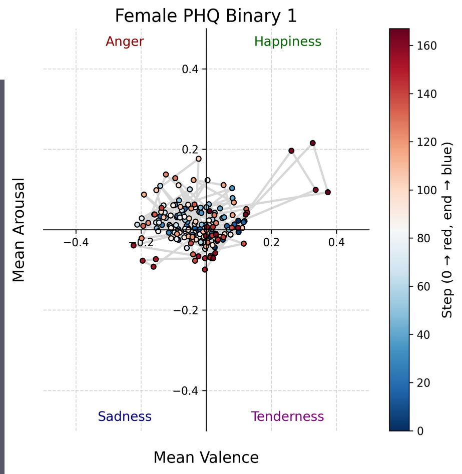

Female PHQ Binary 0
Male PHQ Binary 0
Female PHQ Binary 1
Male PHQ Binary 1
Valence
Arousal
Mean Score
Step

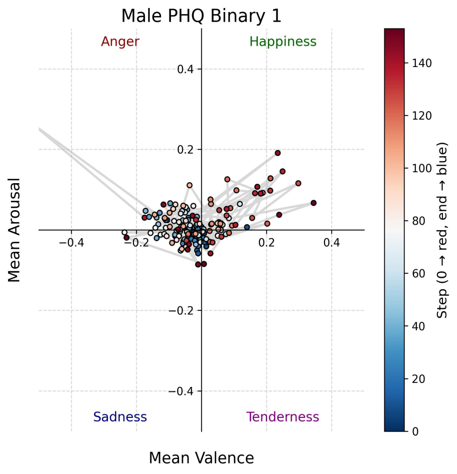
Visualizing Valence & Arousal Trajectories to Reveal Clear Affective Patterns



# of male participants: 43
# of female participants: 13







Results

Insert

1) MAE & RMSE scores across all models (baseline vs +VA)

Background on Evaluation Criteria

- Literature gold-standard
- We chose to use foundation models for the prediction (o4-mini)

Next Steps/Further Research

Agentic vs. Clinical

Agentic

Context (chat history):

- Uses current chat window
- Prior messages are text inputs, rather than a structured history.

Reasoning:

- Investigate user through structured prompting.

Clinical

Context (real time interaction):

- Interprets context across sessions
- Aggregate mood & risks => behavioral trends over time.

Reasoning:

- Investigates patient through psychometrics (eg: PHQ, SAM)
- Result informs treatment plan

2. Prediction of depressive patients

Hypothesis:

labelling DAIC-WOZ transcripts with their valence-arousal scores could improve depression prediction

Our approach:

DAIC-WOZ with or without predicted VA → 4 models → Predicted PHQ-8

- Base GPT-o4-mini
- GPT-o4-mini with chain-of-thought prompting
- GPT-o4-mini with supervised fine-tuning

Evaluation against true PHQ-8 using Mean Absolute Error (MAE)

	Base GPT-o4-mini	GPT-o4-mini with CoT prompting	GPT-o4-mini with SFT
DAIC-WOZ transcripts	4.00	3.66	
DAIC-WOZ transcripts + predicted VA	3.86	3.21	

Published models

Source	MAE (PHQ-8)
Lau et al 2023	3.80
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Need #1: Continuous Mental Health Monitoring

- 31% of adults who have never been diagnosed still screen positive for moderate to severe depression [1].

- Adding routine depression screening to annual check-ups improves treatment and cuts psychiatric hospitalizations [2].

- Passive digital clues (from language to phone sensors) can flag worsening mood in real time, enabling just-in-time LLM or clinician intervention [3].

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Need #2: Scalable Way to Engage with Psychiatrists

[1]: https://pmc.ncbi.nlm.nih.gov/articles/PMC9470500/
[2]: https://pmc.ncbi.nlm.nih.gov/articles/PMC10701157/
[3]: https://mhealth.jmir.org/2024/1/e40689

Medical Basis: The Need for Continuous Mental Health Monitoring

- 31% of adults who have never been diagnosed still screen positive for moderate to severe depression [1].

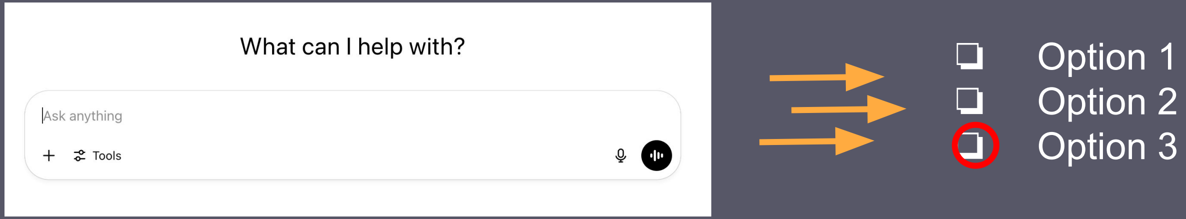
- Adding routine depression screening to annual check-ups improves treatment and cuts psychiatric hospitalizations [2].

Proposed Digital Solution: LLMs to understand emotional state from user text

- Find some statistic on the inadequate accuracy of current problems/how they’re not quantitative (aka technical basis behind our+VA model)

[1]: https://pmc.ncbi.nlm.nih.gov/articles/PMC9470500/
[2]: https://pmc.ncbi.nlm.nih.gov/articles/PMC10701157/
[3]: https://mhealth.jmir.org/2024/1/e40689

Finetuning a GPT / other GPTs (think CoT vs Sys promp)



Need: Scalable clinician capacity

US psychiatrists are fully booked [1]:

- <20% available to see new patients
- 67 day in-person wait time

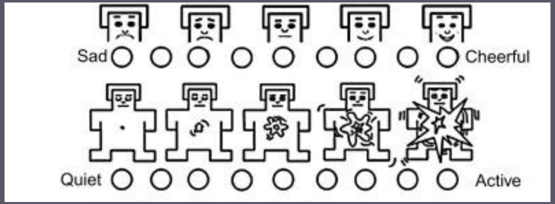
Depression goes untreated [2]:

- LIC depression treatment rate = 16.8%

Guiding Question:

- Is there a foundational psychometric that would allow us to pick up on mental health cues through text?

Solution: Valence-arousal as a psychometric signal

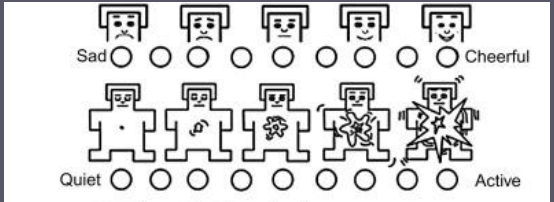


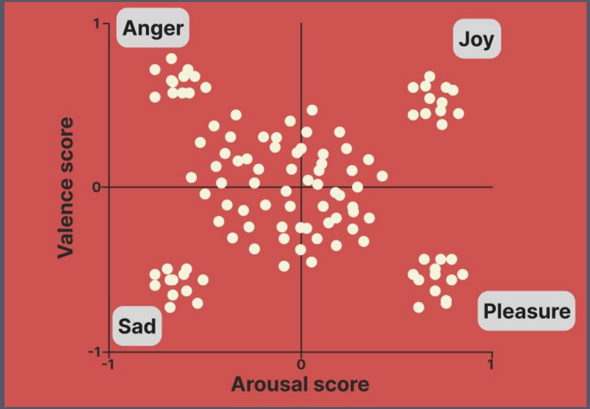
[1]: https://doi.org/10.1016/j.genhosppsych.2023.05.012
[2]: https://pmc.ncbi.nlm.nih.gov/articles/PMC10701157/

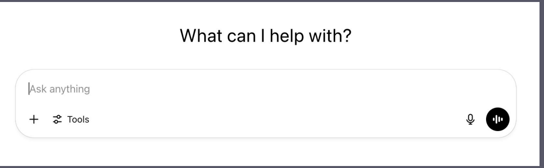
Method: Psychiatric Reasoning

Model + VA => PHQ-8?

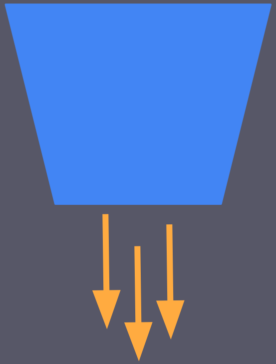
Roberta and EmoBank







{valence: -1, arousal: 1} + prompt



- ❏ Option 1
- ❏ Option 2
- ❏ Option 3